
Rémy Labarre

Hands-on projects



Motivation

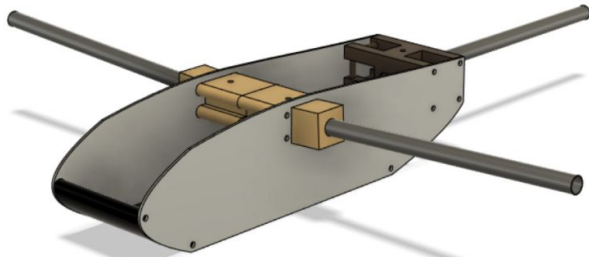
- Mechatronics & robotics education
 - Strong foundation in control systems, sensors, and embedded systems.
- Practical experience with UAVs and robots
 - Projects and AeroTech SeaTech
 - Cranfield UAV oriented master
 - A1, A2 and A3 drone pilot license(EASA / CAA)
- Teamwork & project management
 - Experience from my start-up project and technical collaborations



Dassault UAV Challenge

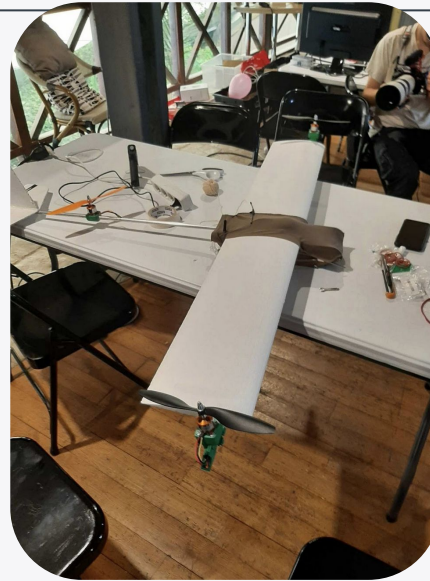
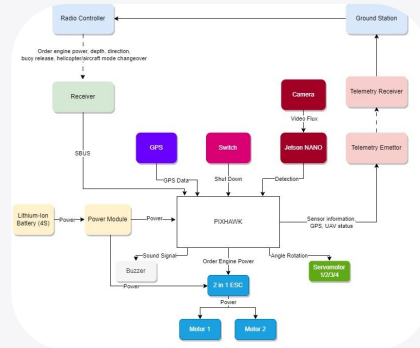
- First steps into UAVs, discovery of the Ardupilot firmware
- Creation of a VTOL tricopter tilt rotor (two motor are vectored for a vertical takeoff and correcting the yaw motion)
- Team ended in the 5th place

Video of the challenge [here](#)

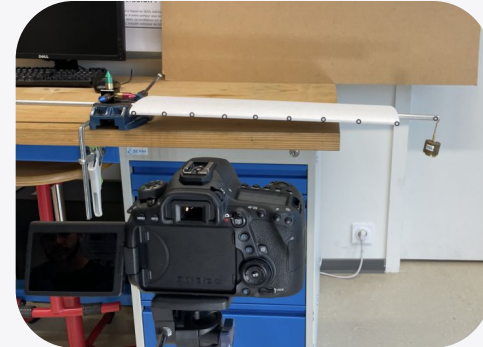


Airframe CAD

Wiring diagram



The UAV

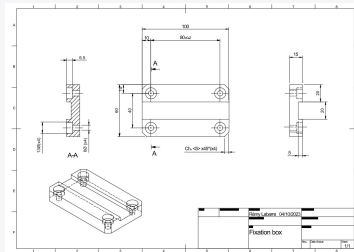


Wing bending tests

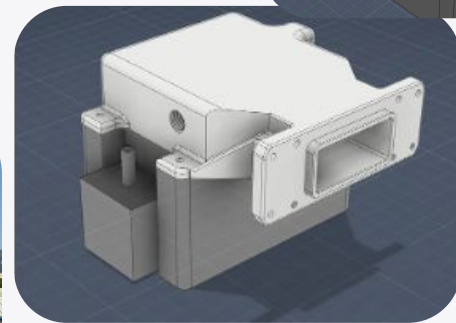
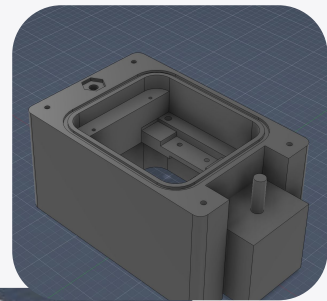
Marine Tech Internship

Robotic technician Internship

- Development of a waterproof enclosure for the integration of an IR camera, an on-board AIS reader and a lidar
- Tests of the AIS receiver
- Integration with the Ardupilot firmware
- Tested a 4G LTE communication system



Fixation Drawings



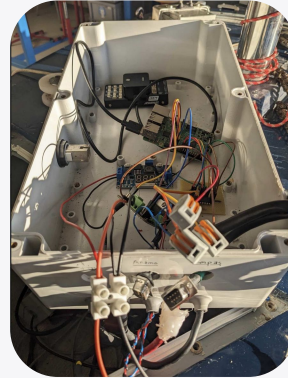
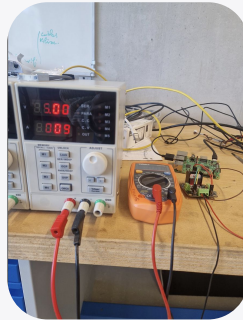
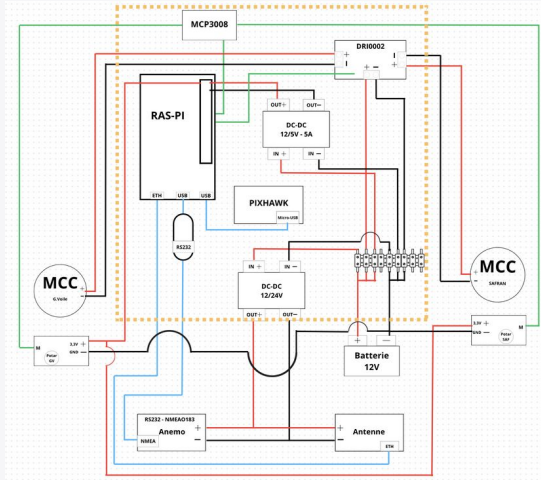
3D housing

Housing integrated on the drone and being field tested

Project Marius

Group project, refurbishment of an old autonomous sailboat to make it up to date:

- Took the continuity of an already begun project
- Control of the different motors to control the sail and the rudder
- Integration and reading of an anemometer
- Video [here](#)



Wiring scheme and tests,
Translation : MMC → DC motor

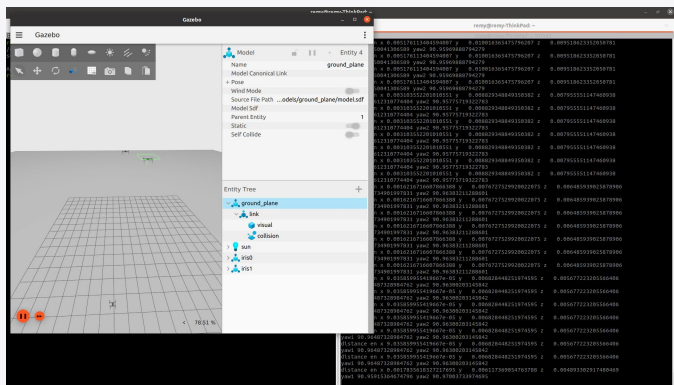
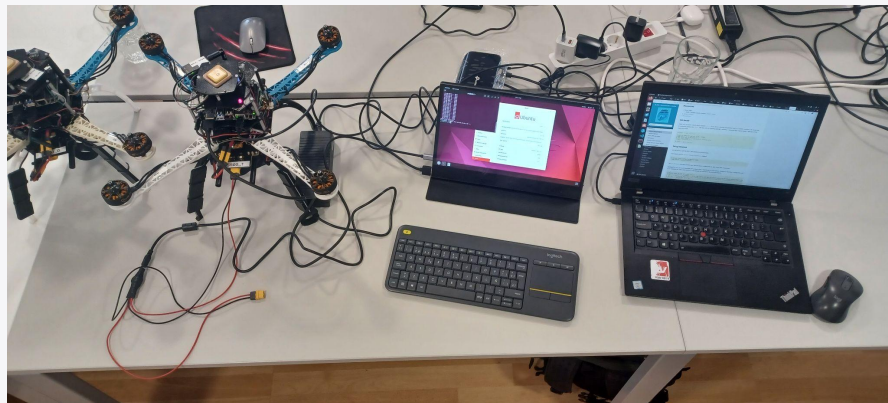


The sailboat

Beyond Vision Internship

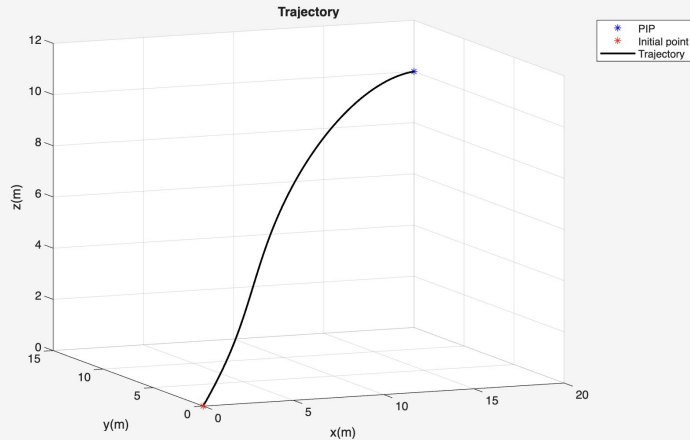
Simulation and application of a swarm of drones
(2 for proof of concept)

- Simulation Gazebo + Ardupilot SITL
- Build of the 2 tests drones (S500 frame + cube orange + Jetson Nano)
- Proved the concept by simulation testing
- Successfully test field-proved the solution
- Video of the tests [here](#)

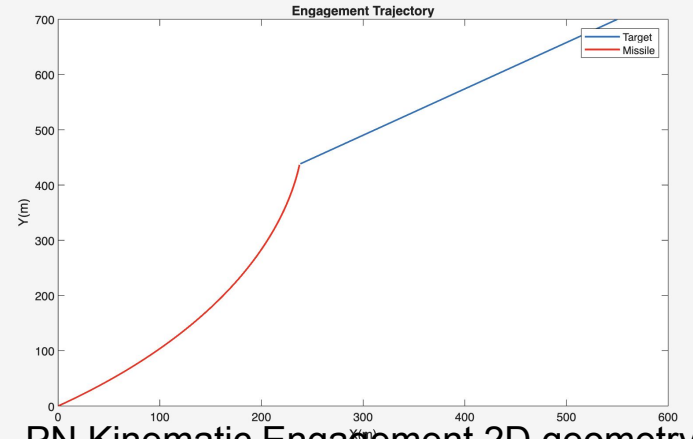


GNC projects

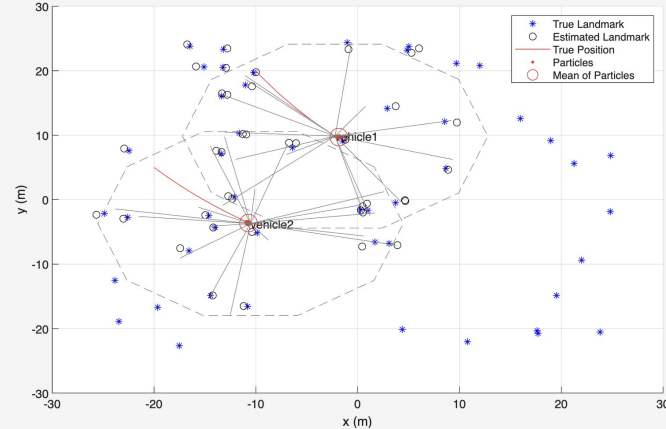
- Dublin Path and Carrot chasing algorithm
- Rapidly exploring Random Tree (RRT) Algorithm
- Generalised Vector Explicit (GENEX)
- Proportional navigation
- SLAM + EKF



GENEX Interception



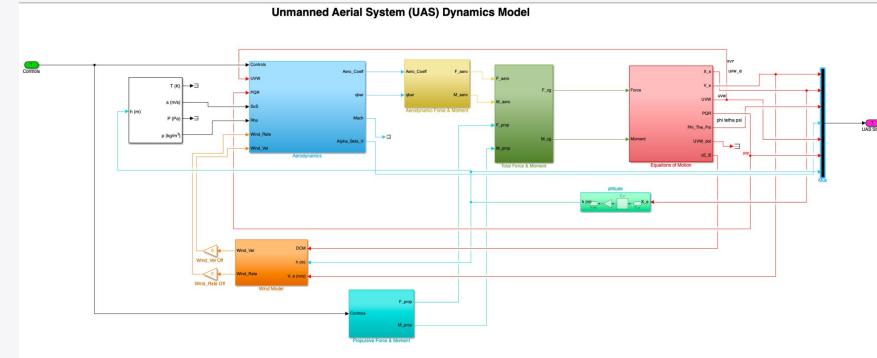
PN Kinematic Engagement 2D geometry



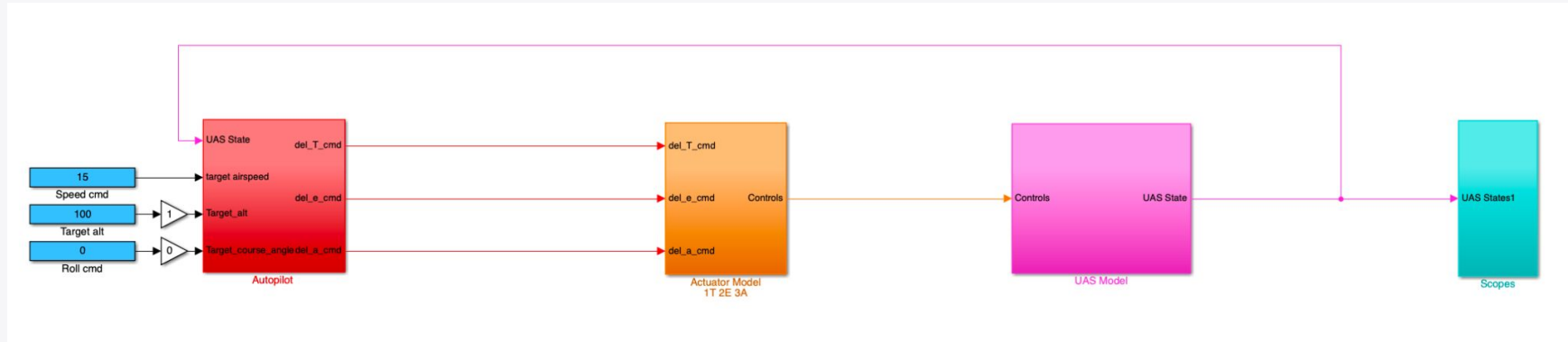
SLAM algorithm for two UAVs

UAS Dynamic and Control

- Build a UAV simulation model
- Analyze and compare its flight behavior
- Do testing, verification, and validation to prove the model is correct and realistic



UAS Dynamics Model

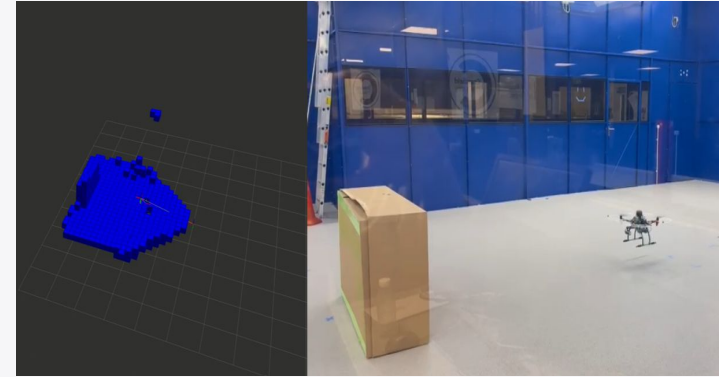


UAS Non-linear Simulation Model

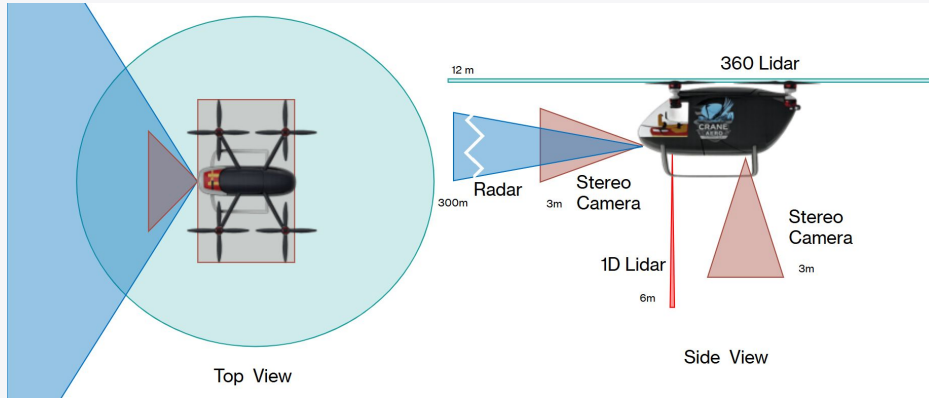
Autonomous Navigation in GNSS denied environment

Group project :

- V&V iteration process
- Choice, positionment and filtering of the sensor suite
- Mapping of the Environment through SLAM
- Running of a path planning algorithm in this Environment
- Integration with a ROS2 workflow + Gazebo testing
- Lab tested the solution



Lab testing the solution



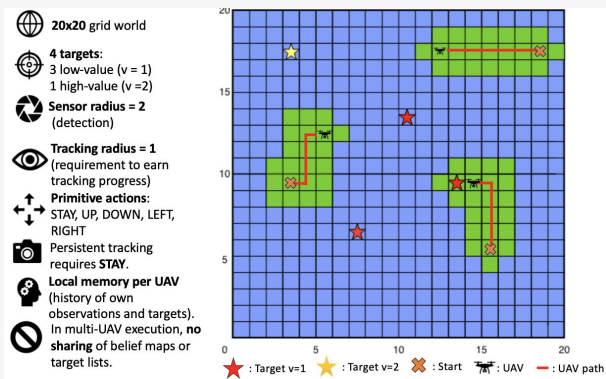
Sensor overview



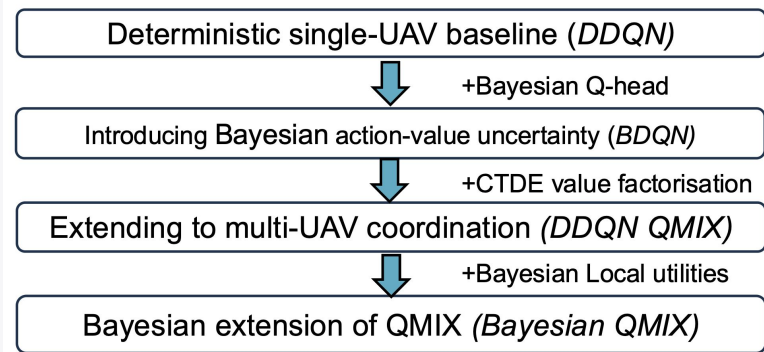
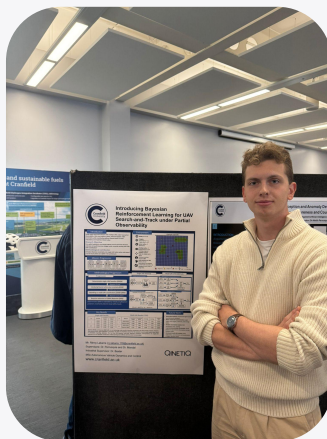
Sensor positioning on a S500 frame

Master Thesis : Multi-agent Reinforcement Learning for search and Track missions

- Investigated Bayesian reinforcement learning for UAV search-and-track under partial observability.
- Developed Bayesian extensions of DDQN and QMIX to model uncertainty in action values.



Episode Overview



Methodological Progression

Teamwork & Collaboration Experience

- Technical competitions (AeroTech SeaTech)
 - Team management and problem-solving
- Start-up project (friendship app)
 - Project management, marketing
- Students projects :
 - Turtlebot (2024-25)
 - Jammer detection with EKF filtering(2026)
- EDTH Hackathon:
 - Maintained drone-swarm positioning in GNSS-denied environments using LoRa RSSI and an embedded EKF, video [here](#)



Marius, autonomous boat project

Association implication

- Drone building association AeroTech SeaTech
 - Drone construction, fund raising
- Students drone flying association :
 - Part of the Cranfield Drone Flying Club, Fpv flying



Cranfield Drone Flying club session

More about me : www.linkedin.com/in/r  my-labarre